Project: Usable Key Management (Part 1)

Overview

In this project, you will be exposed to TLS client authentication, also called mutual TLS or mTLS. This will give you first-hand experience with a modern real-world cryptosystem.

Requirements

To use the pass-off server, you need to authenticate with TLS client authentication. The client (your browser) sends a certificate to the server together with a digital signature produced using the certificate's private key. The server checks that the certificate is signed by a trusted source and that the signature is valid.

To complete the first part of the project and gain access to the pass-off server, follow these steps:

1. Create your cryptographic key pair.
2. Obtain a signed certificate for your key pair from [anonymized institutional site].
3. Register the signed certificate and your private key with your browser.

After successfully completing these steps, you can log into the pass-off website [anonymized institutional site].

You will know everything is working when you can visit this website without receiving an error. After you have completed the above steps and gained access to the pass-off server, write a report describing your experience. This report should include the following details:

1. How long did it take you to complete this project?
2. What steps did you take to complete the project?
    - Also, include details about anything you attempted that ultimately did not work.
    - Include screenshots if you think that would be helpful.
3. Describe what information sources you used and how helpful (or unhelpful) they were.
4. Identify the tools you used.
    - Also, include details about any tools you ultimately abandoned.
    - Describe what went well with these tools and what was challenging.

Reflection Questions

5. Answer the following questions:

    - What were the easiest one or two steps in setting up TLS client authentication and why were they the easiest?

- What were the hardest one or two steps in setting up TLS client authentication and why were they the hardest?
- What would you change about the setup process, if anything?

After-Scenario Questionnaire (ASQ)

6. Indicate how much you agree with the following statements on a scale of 1 (strongly disagree) to 7 (strongly agree):

   - Overall, I am satisfied with the ease of setting up TLS client authentication.
   - Overall, I am satisfied with the amount of time it took to set up TLS client authentication.
   - Overall, I am satisfied with the support information (online help, messages, documentation) I found when setting up TLS client authentication.

System Usability Scale (SUS)

7. Indicate how much you agree with the following statements on a scale of 1 (strongly agree) to 5 (strongly disagree). **Note that this scale is different than the previous scale.**

   - I think that I would have no problem setting up TLS client authentication frequently.
   - I found TLS client authentication unnecessarily complex to set up.
   - I thought that setting up TLS client authentication was easy.
   - I think that I would need help from technical support staff to set up TLS client authentication in the future.
   - I found the various functions for setting up TLS client authentication to be well integrated.
   - I thought there was too much inconsistency in setting up TLS client authentication.
   - I would imagine that most people would learn to set up TLS client authentication very quickly.
   - I found setting up TLS client authentication to be very cumbersome.
   - I felt very confident setting up TLS client authentication.
   - I needed to learn many things before I could get going with setting up TLS client authentication.

8. Lastly, provide any other feedback you have about the project or about setting up TLS client authentication.

Getting Started and Getting Help

As the purpose of this project is to give you experience with a real-world cryptosystem, neither the instructor nor the TAs will tell you how to complete these tasks. **However, you are free to use any online information source you want.** You are also free to use whatever tools you wish to complete this project.

For the learning purposes of this project, avoid asking other students how to complete the project. We recognize that at times the lack of guidance may be frustrating. That is by design. The purpose of this project is to give you experience with using a cryptosystem as they are deployed in the real world.

If after ninety minutes (1.5 hours) you feel stuck on this project, reach out to the TAs for some tips. If after three hours you haven't completed this project, let the TAs know and they will help you complete it so that you can have access to the grading server. In either of the two proceeding cases, mention what help you got from the TAs when writing your report and you can still receive full credit.

Grading Rubric

- **25 points**: for a well-written description of how you set up client authentication. This must include the steps you took as well as the tools and information sources you used (whether successfully or unsuccessfully).
- **15 points**: for answering the three questions about what was easiest and hardest about setting up client authentication and what you would change.
- **5 points**: for answering the SUS questions.
- **5 points**: for answering the ASQ questions.

Total: 50 points